

Computing Limits Analytically

In these sessions¹⁰ we are going to explore the methods to compute limits using algebra, instead of arithmetic, as a preamble to enter into derivatives. We are going to explore 4 main situations: direct substitution, limits for rational, radicals, and at infinity. There is an extra topic, which is trigonometric limits; however, depending on the time, we are only going to mention them or show how to compute the values.

¹⁰ January 27

Direct Substitution

This technique mostly works for any polynomial. For example,

$$\begin{aligned}\lim_{x \rightarrow 3} (2x + 1) &= 2(3) + 1 \\ &= 7.\end{aligned}$$

If you get a function that arrive into situations like $0/0$, $a/0$, we can use the L'Hopital's Rule, which if we have time we are going to cover, if not, then now you know the existence of this algebraic tool.

Rational functions

A common case in which you get into situations like $0/0$ and $a/0$ are the rational functions $\frac{P(x)}{Q(x)}$. This situation emerges when the polynomials $P(x)$ and $Q(x)$ have roots in common, or at the roots of $Q(x)$.

When applying the direct substitution method gives,

$$\lim_{x \rightarrow r} \frac{P(x)}{Q(x)} = \frac{P(r)}{0},$$

then the limit is $\pm\infty$. To identify if it is $+\infty$ or $-\infty$, we need to check the sign of the quotient.

On the other hand, when the direct substitution method arrives to,

$$\lim_{x \rightarrow r} \frac{P(x)}{Q(x)} = \frac{0}{0},$$

then we need to make algebraic manipulation. The main strategy is to factor and cancel common terms, and then apply the direct substitution method,

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} \\ &= \lim_{x \rightarrow 2} (x + 2) \\ &= 2 + 2 \\ &= 4\end{aligned}$$

Remember that for the example, the function it is not defined at $x = 2$, however we can get the exact value of this function by using the idea of limit.

Class Activity Compute the limits of the following functions by using the analytical methods seen in this session.

$$\begin{aligned}\lim_{x \rightarrow 1/2} (4x^2 - 6x + 5) \\ \lim_{x \rightarrow 1} \left(\frac{x}{x^4 + 4} \right) \\ \lim_{x \rightarrow 0} \left(\frac{x}{x^2 + x} \right)\end{aligned}$$